

Design and Control of a Quadrotor Unmanned Aerial Vehicle

A Model-Based Approach Using SolidWorks and MATLAB/Simulink

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1. Introduction

1.1 Overview of Unmanned Aerial Vehicles (UAVs)

Unmanned Aerial Vehicles (UAVs), commonly known as drones, are aircraft systems that operate without an onboard human pilot. These vehicles may be remotely controlled or capable of autonomous flight using onboard sensors, processors, and control algorithms. In recent years, UAVs have gained significant prominence due to advancements in lightweight materials, embedded electronics, sensing technologies, and computational power. These developments have substantially improved the efficiency, precision, and reliability of UAV operations.

UAVs are widely utilized across civil, commercial, and defense sectors. In surveillance and reconnaissance missions, UAVs provide real-time aerial monitoring while reducing risks to human operators. In mapping and surveying applications, drones equipped with cameras and LiDAR sensors enable high-resolution terrain mapping and three-dimensional reconstruction. The logistics sector has also seen increasing interest in UAV-based delivery systems for rapid and cost-effective transportation of lightweight payloads. Furthermore, UAVs play an important role in disaster management, agricultural monitoring, environmental studies, scientific research, and infrastructure inspection, where accessibility and safety are critical considerations.

Among the various UAV configurations, multirotor platforms—particularly quadrotor drones—are extensively used due to their mechanical simplicity, vertical take-off and landing (VTOL) capability, hovering ability, and high maneuverability.

1.2 Motivation for Designing a Quadrotor Drone

A quadrotor UAV consists of four rotors arranged symmetrically about the center of mass. This configuration provides an effective balance between mechanical simplicity and control authority. Unlike fixed-wing UAVs, quadrotors do not require runways for takeoff or landing, making them suitable for operation in confined and urban environments. Additionally, the use of fixed-pitch propellers simplifies the mechanical design by shifting complexity from hardware to software-based control algorithms.

The motivation for selecting a quadrotor platform for this project arises from its relevance in modern UAV research and its suitability for studying multidisciplinary engineering concepts.

Quadrotors serve as an excellent platform for understanding rigid-body dynamics, nonlinear system modeling, feedback control design, and simulation-based validation. From an educational perspective, the quadrotor UAV enables the integration of mechanical design, aerodynamics, propulsion, sensing, and control systems.

Moreover, quadrotor systems are inherently unstable, making their control a challenging problem. Designing a stable and robust controller for attitude and position regulation requires a strong understanding of system dynamics and control theory, thereby contributing significantly to the development of practical aerospace and control engineering skills.

1.3 Scope of the Project

The scope of this project includes both mechanical design and control system development, with an emphasis on simulation-based modeling and validation.

1.3.1 Mechanical Design Using SolidWorks

The mechanical design of the quadrotor UAV is carried out using SolidWorks, a professional computer-aided design (CAD) software. The drone frame is designed to be lightweight while maintaining sufficient structural strength to withstand aerodynamic loads, motor thrust forces, and minor impact stresses. Key components such as the arms, central frame, motor mounts, and payload support structures are modeled with careful consideration of symmetry, mass distribution, and manufacturability.

The SolidWorks model allows accurate estimation of physical parameters such as total mass, center of gravity, and moments of inertia, which are critical inputs for dynamic modeling and control system design. The CAD model also provides a foundation for future modifications, fabrication, and experimental testing.

1.3.2 Dynamic Modeling and Control Using MATLAB/Simulink

The dynamic behavior of the quadrotor UAV is modeled using MATLAB and Simulink. The UAV is represented as a rigid body with six degrees of freedom, consisting of three translational and three rotational motions. The mathematical model is derived using Newton–Euler equations, capturing the nonlinear coupling between rotor thrusts and the resulting translational and rotational dynamics.

A control system is developed in Simulink to stabilize the UAV and achieve desired flight behavior. The control architecture typically consists of an inner-loop attitude controller (roll, pitch, and yaw) and an outer-loop position and altitude controller. Controllers such as PID are implemented to regulate motion and reject disturbances.

1.4 Objectives of the Project

The primary objectives of this project are as follows:

- Design a structurally robust quadrotor drone frame using SolidWorks with proper mass distribution and minimal weight.
- Develop an accurate dynamic model and a stable control system for attitude and position control.

- Validate the control system performance through MATLAB/Simulink simulations using key performance metrics.

2. Mechanical Design Using SolidWorks

2.1 Conceptual Design

An X-type quadrotor configuration with four equally spaced rotors arranged symmetrically about the center of mass is selected for this project. This configuration offers an optimal balance between mechanical simplicity, control effectiveness, and payload capability.

The conceptual layout emphasizes geometric symmetry, efficient component placement, and ease of assembly and maintenance.

2.1.1 Component Placement

- **Motors:** Mounted at the ends of the arms, equidistant from the center.
- **ESCs:** Positioned close to motors to minimize wiring losses.
- **Battery:** Placed near the center of gravity to enhance stability.
- **Flight Controller:** Located at the geometric center to reduce sensor disturbances.

2.1.2 Symmetry and Center of Gravity

Maintaining symmetry about the principal axes ensures that the center of gravity coincides with the geometric center of the quadrotor, which is essential for stable flight and reduced control effort.

2.2 Detailed CAD Modeling

The quadrotor is modeled in SolidWorks using a modular design approach.

- **Central Frame:** Supports avionics and power components.
- **Arms:** Designed to withstand bending and torsional loads.
- **Motor Mounts:** Ensure accurate alignment of thrust vectors.
- **Landing Gear:** Protects the structure during takeoff and landing.

Assembly constraints are applied to ensure symmetry and alignment, and interference checks are performed.

2.3 Material Selection

Material selection is based on strength-to-weight ratio, stiffness, and manufacturability.



Figure 1: Drone SolidWorks

2.3.1 Materials Considered

- Aluminum alloys
- Carbon fiber composites
- PLA/ABS for rapid prototyping

Lightweight materials with high structural performance are selected for critical components.

2.4 Mass Properties Analysis

SolidWorks mass property tools are used to compute total mass, center of gravity, and moments of inertia about roll, pitch, and yaw axes. These parameters are directly incorporated into the MATLAB/Simulink dynamic model.

2.5 Structural Integrity Considerations

Structural design accounts for thrust loads, maneuver-induced stresses, and vibration effects. A conservative factor of safety is applied to ensure reliable operation under nominal and moderately adverse conditions.

3. Mathematical Modeling of the Drone

The quadrotor UAV is modeled as a rigid body with six degrees of freedom.

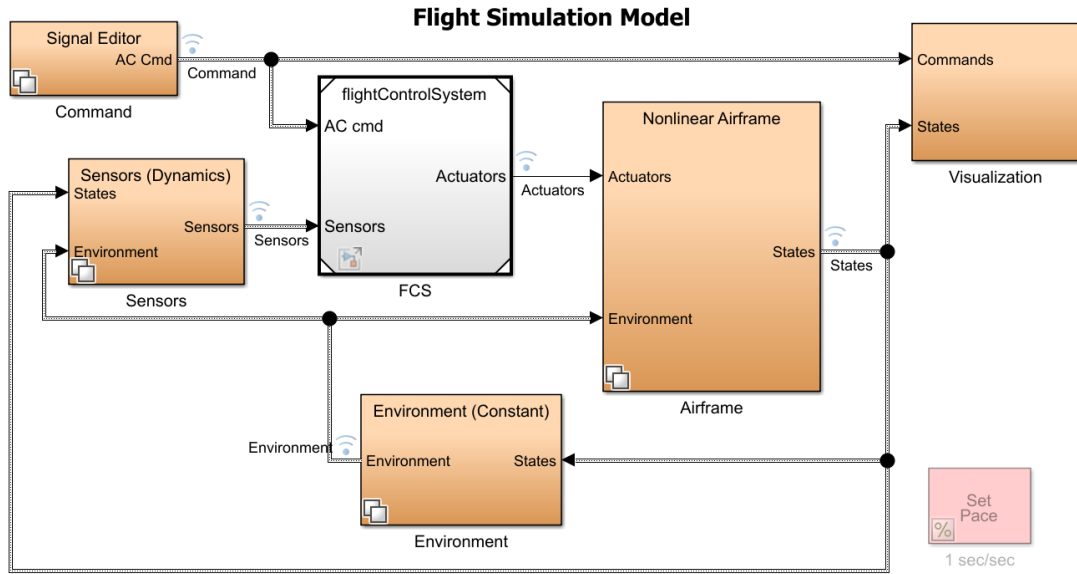


Figure 2: Drone Control Diagram

3.1 Coordinate Frames

Two reference frames are defined: an inertial (Earth-fixed) frame and a body-fixed frame attached to the UAV. Euler angles (roll, pitch, and yaw) are used to represent orientation.

3.2 Forces and Moments

The model includes thrust forces generated by the rotors, gravitational force, aerodynamic drag, and control moments generated by differential thrust and motor reaction torques.

3.3 Equations of Motion

Translational dynamics are derived using Newton's second law, while rotational dynamics are described using Euler's rigid-body equations. Simplifying assumptions such as rigid-body behavior, symmetric mass distribution, and small-angle approximations are applied.

4. Control System Design Using MATLAB/Simulink

4.1 Control Objectives

The control system aims to achieve attitude stabilization, altitude control, position tracking, and disturbance rejection.

4.2 Control Architecture

A cascaded control architecture is adopted, consisting of an inner-loop attitude controller and an outer-loop position and altitude controller.

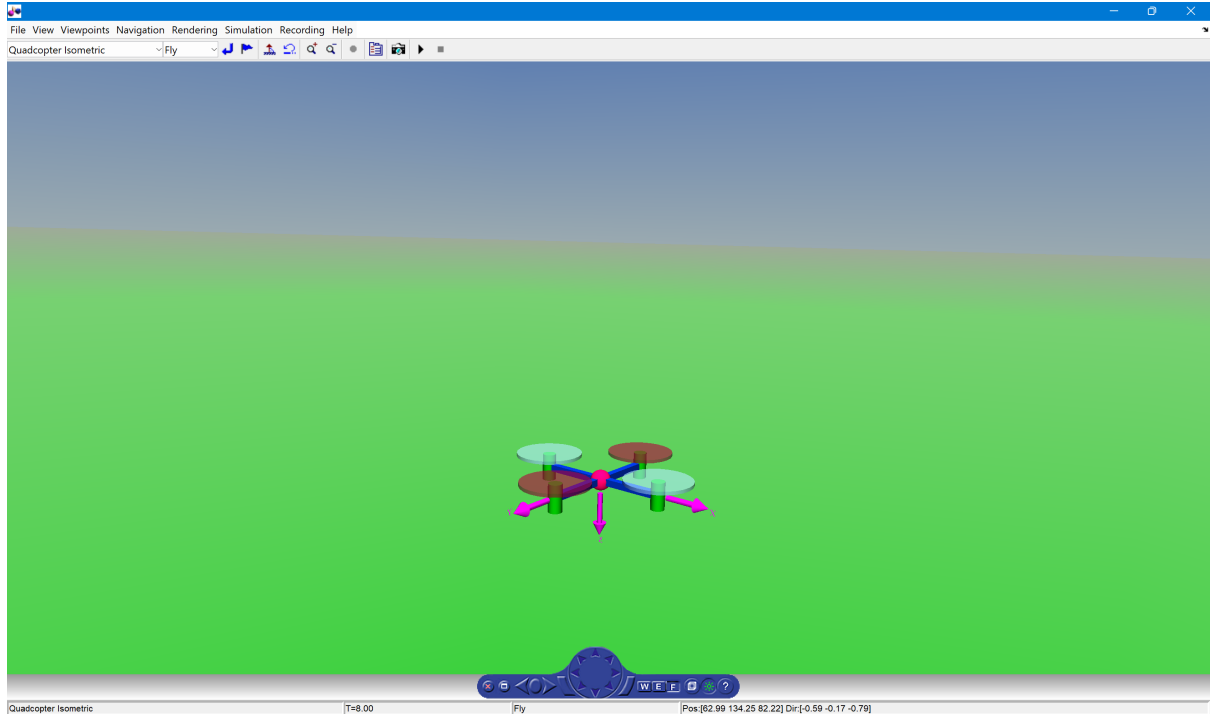


Figure 3: Drone - Simulink

4.3 Controller Selection

PID controllers are implemented for roll, pitch, yaw, and altitude control due to their simplicity, robustness, and widespread industrial usage.

4.4 Simulink Model Development

The Simulink model includes nonlinear quadrotor dynamics, sensor models, PID controllers, and simplified actuator models. Parameters obtained from SolidWorks are directly integrated into the simulation.

4.5 Controller Tuning

Controller gains are tuned using manual adjustment, classical tuning rules, and simulation-based refinement based on rise time, settling time, overshoot, and steady-state error.

5. Conclusion

This project presented the complete design, modeling, and control development of a quadrotor UAV using a model-based engineering approach. Mechanical design was successfully carried out in SolidWorks with careful consideration of structural integrity, symmetry, and mass distribution. A comprehensive mathematical model was developed and integrated with a cascaded PID-based control system in MATLAB/Simulink.

Simulation results demonstrated stable flight behavior, effective disturbance rejection, and satisfactory tracking performance. The methodologies adopted in this project reflect real-world

UAV development practices and provide a strong foundation for future extensions such as hardware implementation, advanced control strategies, autonomous navigation, and experimental flight testing.